ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. AI-DS ENGINEERING — SEMESTER 5

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Explain Supervised, Unsupervised, and Reinforcement Learning paradigms. Bias-Variance Tradeoff. [14 Marks]

[OR]

Q.1 (Alt) Derive Ordinary Least Squares (OLS) for Simple Linear Regression and Gradient Descent update rule. [14 Marks]

--- UNIT II ---

Q.2 Explain Logistic Regression for binary classification. Derive Sigmoid cost function and Cross-Entropy loss. [14 Marks]

[OR]

Q.2 (Alt) Explain Decision Tree splitting criteria: Information Gain (Entropy), Gini Impurity, and CART algorithm. [14 Marks]

--- UNIT III ---

Q.3 Explain Support Vector Machines (SVM): Maximum Margin Hyperplane and Kernel Trick (RBF, Polynomial). [14 Marks]

[OR]

Q.3 (Alt) Describe Ensemble Learning: Bagging (Random Forest) vs Boosting (AdaBoost, Gradient Boosting, XGBoost). [14 Marks]

--- UNIT IV ---

Q.4 Explain K-Means Clustering algorithm. How does Elbow Method determine optimal K? [14 Marks]

[OR]

Q.4 (Alt) Explain Principal Component Analysis (PCA) for Dimensionality Reduction using Eigenvalue decomposition. [14 Marks]

--- UNIT V ---

Q.5 Explain Evaluation Metrics: Precision, Recall, F1-Score, ROC-AUC Curve, and Confusion Matrix. [14 Marks]

[OR]

Q.5 (Alt) Describe K-Nearest Neighbors (KNN) algorithm and Naive Bayes Classifier with Laplace smoothing. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****